

Valerian Fourel

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SUMMARY

Machine Learning Engineer specializing in LLMs, RAG, agentic AI, and production ML systems, with experience taking AI solutions from research to deployment. Combines strong technical depth in **LLM post-training, evaluation, retrieval, and ML engineering** with a proactive, product-oriented mindset and strong collaboration across **research, engineering, and business teams**.

WORK EXPERIENCE

University of Hamburg / Economic AI GmbH

Hamburg, Germany

Machine Learning Engineer

Aug 2025 - Present

- Architected a **reproducible LLM evaluation and reranking framework** with versioned prompt specifications, evaluation sets, provider abstractions, caching, schema validation, and failure-safe logging; tested system-prompt variations and randomized candidate ordering to measure accuracy, consistency, and ranking robustness at cluster scale.
- Developed **LLM post-training and distillation workflows** using supervision from 70B+ models to fine-tune smaller models for agentic search, benchmarking capability retention and inference efficiency to select models for different tasks.
- Developed and optimized **50K+ causal machine learning workflows** using **PySpark and SQL on AWS Databricks**, **reducing runtime by 4×** and supporting scalable experimentation for **8+ engineers**.

Max Planck Institute for Intelligent Systems

Tubingen, Germany

Research Assistant

Aug 2023 - Aug 2025

- **Engineered multimodal data and embedding pipelines** across **50,000+ motion sequences**, processing textual, visual, and 3D representations with **Python and PyTorch** for state-of-the-art human body modelling (SMPL-X).
- **Fine-tuned generative and multimodal foundation models**, including **Stable Diffusion**, for photorealistic facial reconstruction and affect modelling across **500K images and 8 emotion categories**, improving task accuracy by **25%** through perceptual-loss optimization.
- **Containerized ML research workflows with Docker**, standardizing reproducible model environments for **10+ researchers** and reducing simulation and environment-sharing time by **90%**.

Soil Science Lab - University of Tübingen

Tubingen, Germany

Data Science Intern

Jan 2023 - Aug 2023

- Built a **JAX-based data pipeline** to preprocess geospatial data and validate feature extraction across multiple predictive modelling tasks.
- **Analyzed 16K soil samples across Bavaria**, developing and validating predictive models using geospatial and environmental data.
- **Publication:** [Fourel et al., *Geoderma* \(2026\)](#) — compact CNN–Transformer models for spatially robust soil organic carbon mapping.

SELECTED AI PROJECT

SeoulDoc.io

Seoul, Korea

Creator & AI Engineer

Jan 2026 - Present

- **Designed and publicly deployed an end-to-end RAG and agentic search application** covering 8,484 healthcare facilities and 1.25M review embeddings, combining structured enterprise-style data, vector retrieval, ranking, prompt orchestration, and LLM reasoning.
- Built **tool-calling agent workflows** integrating **vector search and LLM APIs** for information retrieval and recommendation generation.
- Implemented evaluation, safety constraints, and deployment workflows for a high-stakes medical-information domain.

EDUCATION

Eberhard Karls University Tübingen

Tubingen, Germany

Master of Science - Quantitative Data Science - Machine Learning Specialization

McGill University

Montreal, Canada

Bachelor of Arts - Computer Science and Economics

SKILLS & QUALIFICATIONS

Skills: Python (PyTorch, JAX, PySpark) · LLM Post-Training & Evaluation · RAG & Agentic AI · Tool Calling · Docker · CI/CD · SQL · Databricks · AWS · Causal Machine Learning

Interests: Equities & Technology Investing · Music (1950s to 1980s) · Board Games · Functional Training · Learning Korean

Languages: German (C1) · English (C2) · French (Native Speaker) · Korean (A2)